Field-Scale Well Construction Timeline Reconstruction from WITSML Daily Drilling Reports: A Complete Operational History of the Equinor Volve Dataset

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Abstract

[TJ: rewrite in your own words]

Daily Drilling Reports (DDRs) are the most complete operational record of how a field's wells were built, but they are rarely converted into structured, validated construction timelines, and no such timeline has been published for an openly available field dataset. This paper reconstructs field-scale well construction timelines for the Equinor Volve dataset from 1,759 WITSML DDRs covering 26 wellbores in 11 well families, and validates the result against the Norwegian Offshore Directorate (NOD) FactPages.

A deterministic, priority-ordered, rule-based classifier assigns each DDR's 24-hour narrative to one of 13 phases. It assigns 1,303 DDRs (74.1%). A stratified audit of 50 records, drawn from the 383 unclassified DDRs that carry a genuine narrative, found 26 (52%) describing construction operations the classifier missed: drilling progress reported without a hole size, non-standard abbreviations, and operations such as fishing and fracturing for which no pattern exists. Scaled to the sampled population, this gives an estimated attainable classification rate of 85.4% (95% confidence interval 82.5 to 88.3%).

Of 24 wellbores with a defensible NOD counterpart, 22 agree exactly on total depth. 15/9-F-15 differs by 15 m, and 15/9-19 S differs by 227 m because the DDR archive does not cover that wellbore's final drilling. All three naming mismatches between the two registers follow the Norwegian designation convention for technical sidetracks. Spud date comparison identified four WITSML data-entry errors, eight definitional offsets, and one NOD entry date that marks a later drilling campaign. The archive also stores placeholder values in 515 fields (331 depth sentinels, 7 zero depths, 176 literal "None" strings and one date sentinel) that a simple null test does not detect.

Sidetrack kick-off depths were recovered for all 15 sidetracks by one rule; where NOD records a kick-off point (12 of 15), the recovered depth agrees within 10 m for 10. The output is an open depth-time-phase dataset of 1,421 records, with per-wellbore durations to total depth, published with the classifier code. It supports days-versus-depth benchmarking and records-based screening of legacy wellbores, and the method transfers to other WITSML 1.4-series drillReport archives. The results show that neither the structured fields nor the national register alone gives a complete well history; the narrative reports are needed to correct both.

1 Introduction

Daily Drilling Reports (DDRs) are the primary operational record of a well construction campaign. Each report logs depth progress, casing and cementing operations, formation encounters and non-productive time. In the WITSML (Wellsite Information Transfer Standard Markup Language) exchange format (Energistics n.d.), a DDR holds structured fields such as measured depth, spud date and casing depths, together with a free-text narrative of the preceding 24 hours of operations. Turning the archive of a field into a construction timeline means assigning every report to a construction phase, and the inconsistent terminology, abbreviations and narrative text that accumulate over a field's multi-decade life make that difficult. This paper reconstructs the timeline of the Equinor Volve field from 1,759 WITSML DDRs covering 26 wellbores. A deterministic, priority-ordered rule-based classifier reads the narrative of every report. A stratified sample of the unclassified remainder shows what the classifier misses, and total depth and spud date are validated for every wellbore against an independent national register, the NOD FactPages (NOD 2026a). The data-quality problems that any other user of the archive will meet are documented along the way. A reader with this text, the openly available Volve dataset (Equinor 2018) and the accompanying repository should be able to reproduce every reported figure.

2 Background and Related Work

Several studies have extracted structured information from the free text of drilling reports. Sidahmed et al. (2015) built a dictionary of rig terms and acronyms with drilling experts and tracked weighted term frequencies over time in the reports of three wells to flag recurring operational concerns. Priyadarshy et al. (2017) presented a framework for predicting the causes of non-productive time from unstructured reports. Hoffimann et al. (2018) applied deep natural language processing to 9,670 daily reports from a field of 303 wells and classified each sentence as an event, a symptom or an action. They noted that technical symbols, abbreviations, misspellings and incomplete sentences make the text difficult to process. Zhang et al. (2022) built a rule-based detector for five unscheduled events (stuck pipe, lost circulation, water influx, gas influx and H2S) in about 128,000 free-text remarks from daily mud reports. The detector was precise but missed events, and its accuracy fell to 65.18% on a larger test set. A hybrid that used its output to train a weakly supervised deep learning model improved detection by more than 20 percentage points.

Karnot et al. (2025) combine change-point detection, semi-supervised natural language processing and sequence mining for root-cause analysis of non-productive time and apply the combination to 701 DDRs from 10 wells. Their Dataset section states that "all analyses used synthetic datasets constructed to emulate typical drilling records". Their sequence-mining section, which they describe as a proof-of-concept analysis, states that "rigorous validation and quantitative performance assessment are left for future work". Their abstract and conclusions nevertheless report an average 18.5% reduction in non-productive time in field applications.

The Volve dataset is an established benchmark for drilling analytics. Tunkiel et al. (2021) derived a reference dataset for rate-of-penetration benchmarking from its drilling data. Commercial analyses of Volve DDRs for subsets of wells have been described informally. To the author's knowledge, no validated field-wide construction timeline from the Volve DDR narratives has been published.

Published DDR work relies on proprietary or synthetic data, which prevents others from reproducing or extending its results. The present work (1) uses a deterministic, auditable classifier that trades model sophistication for transparency of every decision, (2) works exclusively with a real, openly licensed archive, and (3) pairs classification with a stratified audit of the unclassified residual, using a documented sampling frame and a Wilson-score interval (Wilson 1927), and with validation of total depth and spud date against a national register for the complete wellbore population. The audit in this paper measures the weakness that Zhang et al. (2022) report, the events a rule-based classifier misses, directly on the unclassified reports.

3 Data and Methods

3.1 Data Source and Extraction

The 1,759 WITSML 1.4.0.0 XML files of the Volve dataset (21,315,879 bytes, each recorded with a SHA-256 hash) were extracted into a single table with one row per DDR and no duplicate wellbore and date pairs. The rows cover the exploration well 15/9-19 and the platform wells 15/9-F-1, F-4, F-5, F-7, F-9, F-10, F-11, F-12, F-14 and F-15, which make up 26 wellbores in 11 well families. The columns include the source XML filename, the well and wellbore identifiers, the report date, the spud date (WITSML dTimSpud), measured depth (md_m), kick-off depth (md_kickoff_m), last casing depth (md_csg_last_m), operational parameters and the free-text 24-hour narrative (sum24Hr). Every field was kept as a raw string to keep placeholder values visible, and the published code regenerates the table from the raw files.

The md_m field is stored as a string and contains two placeholders, the literal '-999.99' (a missing-value sentinel) and the literal '0'. Both were excluded and the remaining values cast to a numeric type before any depth arithmetic.

Table 1 shows a real construction report as stored in the archive, with the structured fields the pipeline uses and the full free-text summary that the classifier reads. Fig. 1 shows the same report as a page of the original PDF version of the Daily Drilling Report.

Table 1: Example DDR as stored in the archive

Field	Value	Meaning
wellbore	15/9-F-15 D	the wellbore the report belongs to
date_report	2013-11-23	the day the report covers
md (m)	2,961	measured depth reached, along the hole
dTimSpud	2013-11-10	spud date in the report header, the same in every report of the wellbore
md_kickoff	absent	kick-off depth field: empty in this report, as in most reports (Table 3)
md_csg_last (m)	2,612	depth of the last casing shoe
forecast24hr	Drill 12 1/4" hole to section TD at approx. 3300 m MD.	the plan for the next 24 hours
sum24Hr	“Performed FIT to 1,55 sg EMW. Drilled 12 1/4" hole from 2623 m to 2961 m through Ty Fm into Shetland Gp. Bled in LCM mix while drilling Ty Fm. and maintained background concentration throughout section. Built 10 3/4" and 9 5/8" casing stands offline. Racked back 30 stands in derrick.”	free-text summary of the past 24 hours: the text the classifier reads
Phase assigned	12¼ in. and smaller hole (priority 5)	the classifier's result: the 12¼ in. hole matches the pattern of that phase

Source: Equinor Volve dataset (Equinor 2018), file 15_9_F_15_D_2013_11_23.xml. The summary of 51 words and the forecast are quoted in full. The kick-off field is empty in this report, as in most reports (Table 3).



Summary report

Wellbore: 15/9-F-15 D — **Period: 2013-11-22 00:00 - 2013-11-23 00:00**

Status:	normal
Report creation time:	2018-05-03 13:52
Report number:	13
Days Ahead/Behind (+/-):	3.1
Operator:	Statoil
Rig Name:	MÆRSK INSPIRER
Drilling contractor:	Maersk Drilling
Spud Date:	2013-11-10 14:00
Wellbore type:	
Elevation RKB-MSL (m):	54.9
Water depth MSL (m):	91
Tight well:	Y
HPHT:	Y
Temperature ():	
Pressure ():	
Date Well Complete:	2013-12-02

Dist Drilled (m):	338
Penetration rate (m/h):	-999.99
Hole Dia (in):	12.25
Pressure Test Type:	formation integrity test
Formation strength (g/cm3):	1.55
Dia Last Casing ():	

Depth at Kick Off mMD:	
Depth at Kick Off mTVD:	
Depth mMd:	2961
Depth mTVD:	2698
Plug Back Depth mMD:	
Depth at formation strength mMD:	2623
Depth At Formation Strength mTVD:	2468.9
Depth At Last Casing mMD:	2612
Depth At Last Casing mTVD:	2457.4

Summary of activities (24 Hours)

Performed FIT to 1,55 sg EMW.
Drilled 12 1/4" hole from 2623 m to 2961 m through Ty Fm into Shetland Gp.
Bled in LCM mix while drilling Ty Fm. and maintained background concentration throughout section.
Built 10 3/4" and 9 5/8" casing stands offline. Racked back 30 stands in derrick.

Summary of planned activities (24 Hours)

Drill 12 1/4" hole to section TD at approx. 3300 m MD.

Fig. 1: Page of the original Daily Drilling Report for 15/9-F-15 D, 23 November 2013 (Equinor Volve dataset, PDF version). Table 1 shows the same report as stored in the WITSML archive.

3.2 Wellbore Designation

Norwegian wellbore names follow the designation guidelines of the national regulator (NOD 2025). The suffix "S" marks an exploration wellbore planned to be deviated, letters A, B and so on mark sidetracks, and "R" marks a re-entered exploration wellbore. A technical sidetrack, drilled to bypass a mechanical problem such as stuck pipe, is designated T2 for the first and T3 for the next. The Volve WITSML archive uses the technical-sidetrack designations

15/9-19 BT2, 15/9-19 ST2 and 15/9-F-11 T2, and none of them appears as a separate entry in the NOD register for the field. This one convention explains every naming mismatch between the two registers (Section 5.4).

3.3 Phase Classification

The classifier assigns the sum24Hr narrative of each DDR to one of 13 phases: Spud/Conductor, Surface hole and casing, Intermediate drilling, Intermediate casing, 12¼ in. and smaller hole, Liner/Casing run, Cementing, Completion, P&A, Logging/Testing, Workover, Waiting/Downtime and Rig move/Setup. Each phase has a fixed priority from 1 (highest) to 13 (lowest), in the order listed, and a set of case-insensitive regular expressions. A DDR takes the highest-priority phase with a matching pattern, and a narrative that matches no pattern stays Unclassified. Fraction characters such as ½ and ¼, routine in the mixed units of drilling reports, are stored as single special characters that a pattern written as "1/2" does not match. They are converted to that form before matching. The phase names refer to the hole and casing sizes that each pattern group matches. On Volve platform wells the 12¼ in. hole is usually the section above the reservoir. Every regular expression is listed in the repository file src/classifier_v3.py.

Table 2 gives the 13 phases in priority order with representative triggers from the pattern table in the repository (outputs/stage4_priority_rules.csv). The patterns match sizes in the spellings found in the reports, for example 17 1/2 or 17.5, and words in any case. Each phase has further patterns that the table does not show.

Table 2: Phases in priority order with representative triggers

Priority	Phase	Hole and casing sizes (in.)	Example trigger words
1	Spud/Conductor	36, 30	spud, tagged sea bottom
2	Surface hole and casing	26, 20	(size only)
3	Intermediate drilling	17½	(size only)
4	Intermediate casing	13⅜, 10¾, 14	(size only)
5	12¼ in. and smaller hole	12¼, 8½, 6	cut core, coring
6	Liner/Casing run	9⅝, 7	liner, surface tieback
7	Cementing	none	cement, cemented, cementing, bumped plug
8	Completion	none	completion, production packer, packer, tubing
9	P&A	none	EZSV, EVO plug, P&A, abandon
10	Logging/Testing	none	logging, LWD, MWD, VSP
11	Workover	none	workover, coiled tubing, sidetrack, whipstock
12	Waiting/Downtime	none	WOW, waiting on weather, rig maintenance, general maintenance
13	Rig move/Setup	none	skidded rig, positioned rig, mooring, ballast

A report is assigned to the highest-priority phase with a match; a report that matches no pattern is Unclassified.

3.4 Stratified Audit of the Unclassified Residual

The audit estimates how many unclassified DDRs describe construction operations that the pattern set failed to catch. A leak check first reapplied the complete pattern set to every unclassified record and found no record that should have matched. Every unclassified record was then assigned to one of eleven descriptive categories, for example BOP and riser handling, generic trips, production monitoring and empty records, using patterns built from inspection of the unclassified population. BOP and marine riser handling is the largest category (147 of 456 records, 32.2%). The 73 empty records have a sum24Hr field that holds the literal string "None" (Section 4). No text method can classify them, and they were excluded from the sampling frame, which left 383 records.

A stratified sample of 50 records was drawn from the 383. The unclassified reports were split into two strata, an early construction group (reports within 730 days of the wellbore's first report, 296 records) and a later production group (reports more than 730 days after it, 87 records), and a random sample was drawn from each in proportion to its size, 39 records from the construction group and 11 from the production group. The random draw is recorded in the

repository and allows the same sample to be drawn again. The author read each sampled report and labelled it construction if it described work that builds the well (drilling ahead, coring, running or cementing casing, setting plugs, fishing or completion). Reports that described only BOP or riser handling, tripping, waiting or production were labelled non-construction unless they also named a hole or casing size.

3.5 External Validation Against NOD FactPages

Total depth and spud date were compared with the NOD FactPages (NOD 2026a). The Table View export was filtered to the Volve wellbores and matched by exact name wherever possible. Three WITSML wellbores, all technical sidetracks, were matched by depth (15/9-19 BT2 to NOD 15/9-19 B, both 4,250 m; 15/9-19 ST2 to NOD 15/9-19 SR, both 4,641 m; and 15/9-F-11 T2 to NOD 15/9-F-11, 4,562.1 m and 4,562.0 m). The export was retrieved on 24 October 2026. Its rows for the Volve wellbores match an export of 8 September 2026 except for the register's own synchronisation date. All comparisons are against the current register as accessed in October 2026.

In the 15/9-19 series the DDR narratives describe the events behind the designations, and the NOD wellbore histories (NOD 2026b, 2026c, 2026d) are consistent with them. The drillstring in 15/9-19 B became stuck at 3,272 m. The string was backed off and cut, a sidetrack plug was set, and new hole was kicked off from 2,911 m as 15/9-19 BT2, whose first report reads "REPORTING TRANSFERRED FROM 15/9-19B". Well 15/9-19 S followed the same sequence, with stuck pipe at approximately 2,741 m, a backed-off and cut string, sidetrack language in the narrative ("MU BHA FOR SIDE TRACK") and new hole from approximately 1,536 m as 15/9-19 ST2. In both cases NOD records the final total depth against the parent designation.

4 Data Quality

The population of ten WITSML-derived columns was surveyed across all 1,759 DDRs (Table 3).

Table 3: Field population survey (n = 1,759 DDRs)

Column	Populated	Count	Notes
date_report	100.0%	1,759	
spud_date	100.0%	1,759	Includes inherited and erroneous values (Section 5.5)
rig	99.94%	1,758	**1 record missing (1980 sentinel)**
forecast24hr (genuine)	94.14%	1,656	**103 literal "None" strings excluded**
operator	99.94%	1,758	1 record missing
md_csg_last_m	96.76%	1,702	
sum24Hr (genuine narrative)	95.85%	1,686	73 literal "None" strings excluded
drill_complete	91.70%	1,613	
md_m (excluding sentinel and zero)	80.78%	1,421	331 sentinel and 7 zero values excluded
md_kickoff_m	14.10%	248	Free-text recovery required

Percentages are count / 1,759, rounded to two decimal places.

The structured kick-off field is populated in only 14.10% of rows (248 of 1,759), so sidetrack kick-off depths had to be recovered from the narratives and from depth sequences (Section 5.6).

A standard null-or-blank check reports 100% population for sum24Hr, but 73 records hold the literal string "None". The string sits in the source XML, in 73 of the 1,759 files, and is a placeholder of the archive. A null test cannot detect it, and the genuine narrative population is 95.85% (1,686 of 1,759). The same placeholder fills 103 forecast24hr values, which leaves a genuine forecast population of 94.14% (1,656 of 1,759). The archive holds 515 placeholder values across all extracted fields, made up of 331 depth sentinels, 7 zero depths, 176 literal "None" strings (73 narratives and 103 forecasts) and one date sentinel.

A character-level survey of sum24Hr found two non-ASCII characters that matter for classification, the one-half sign (½) in 61 DDRs (82 occurrences) and the one-quarter sign (¼) in 10 DDRs (14 occurrences). Both are normalised

before pattern matching. The raw files declare the ISO-8859-1 character encoding, and reading them with another encoding turns ½ into Â½ (for example 'due to double 3 Â½" DP' for 'due to double 3 ½" DP'). The encoding must therefore be set explicitly. One further anomaly is a single record for 15/9-19 A dated 1 January 1980 (filename 15_9_19_A_1980_01_01.xml, narrative "None", no measured depth), 17.6 years before the first genuine report of the wellbore on 25 July 1997. It is a date sentinel.

5 Results

5.1 Classification Rate

The classifier assigns a phase to 1,303 of the 1,759 DDRs (74.1%) and leaves 456 unclassified (25.9%) (Table 4). The classifier was also checked against an independent earlier extraction of the same archive by the author, with identical results for all 1,759 reports. Fig. 2 shows the classified phases on the depth-versus-time curves of four representative wellbores.

Table 4: Phase distribution

Phase	DDR count	% of 1,759
12¼ in. and smaller hole	256	14.6
Completion	228	13.0
Liner/Casing run	144	8.2
Waiting/Downtime	120	6.8
Cementing	100	5.7
Logging/Testing	93	5.3
Intermediate casing	82	4.7
Intermediate drilling	77	4.4
Surface hole and casing	66	3.8
Spud/Conductor	52	3.0
P&A	41	2.3
Rig move/Setup	38	2.2
Workover	6	0.3
Unclassified	456	25.9
Total	1,759	

Percentages are rounded independently and may not sum to exactly 100.0%.

5.2 Stratified Audit and Attainable Classification Rate

A complete pattern set would classify about 85.4% of the 1,759 reports (95% confidence interval 82.5 to 88.3%). Of the 50 sampled reports, 26 (52%) were labelled construction and 24 (48%) non-construction. The construction group (39 reports) split 23 construction to 16 non-construction, and the production group (11 reports) split 3 to 8. Of the 26 construction reports that the classifier missed, 9 reported genuine drilling without a hole size, 10 described other operations with no corresponding pattern, such as hydraulic fracturing or fishing, and 7 used non-standard terms, such as "CMT" for cement. Scaled to the 383-report frame, the 26 correspond to about 199 reports. The arithmetic, including the stratum-weighted check, is in Appendix B.

No text method can classify the 73 DDRs whose narrative is the literal string "None". They are excluded from the attainable rate, and even a perfect pattern set leaves at least 4.2% of DDRs (73/1,759) unclassified.

5.3 Timeline Dataset

The timeline dataset holds 1,421 rows, one for each DDR with a genuine md_m value. Of the 1,759 DDRs, 331 carry the sentinel and 7 the zero value. Each row gives the wellbore identifier, the report date, the days from the wellbore's first report with a recorded depth, the measured depth, the assigned phase and the source filename. Days are counted

from the first row with a usable depth in each wellbore. Fig. 3 plots report number against measured depth for all 26 wellbores. The report number counts every DDR of a wellbore from its first report (1980 sentinel excluded), with or without a recorded depth, and so matches the reports to TD in Table 8. Only reports with a valid depth are plotted.

5.4 Total Depth Validation

Maximum reconstructed measured depth per wellbore was compared with NOD's total depth for all 26 wellbores (Table 5, Fig. 4).

Table 5: Total depth validation, all 26 wellbores

Wellbore (WITSML)	WITSML max MD (m)	NOD TD (m)	Diff. (m)	Outcome	Note
15/9-19 A	4,131.0	4,131.0	0.0	Exact	
15/9-19 B	3,272.0	n/a	n/a	Unmatched	Parent leg abandoned at stuck pipe; NOD 19 B TD matches 19 BT2
15/9-19 BT2	4,250.0	4,250.0	0.0	Exact	Technical sidetrack, matched to NOD 15/9-19 B
15/9-19 S	3,353.0	3,580.0	227.0	Close	Archive ends before final drilling
15/9-19 ST2	4,641.0	4,641.0	0.0	Exact	Technical sidetrack, matched to NOD 15/9-19 SR
15/9-F-1	3,632.0	3,632.0	0.0	Exact	
15/9-F-1 A	3,682.0	3,682.0	0.0	Exact	
15/9-F-1 B	3,465.0	3,465.0	0.0	Exact	
15/9-F-1 C	4,094.0	4,094.0	0.0	Exact	
15/9-F-10	5,331.0	5,331.0	0.0	Exact	
15/9-F-11	347.0	n/a	n/a	Unmatched	Parent leg; NOD 15/9-F-11 TD matches F-11 T2
15/9-F-11 A	3,762.0	3,762.0	0.0	Exact	
15/9-F-11 B	4,770.0	4,770.0	0.0	Exact	
15/9-F-11 T2	4,562.1	4,562.0	0.1	Exact	Technical sidetrack, matched to NOD 15/9-F-11
15/9-F-12	3,520.0	3,520.0	0.0	Exact	
15/9-F-14	3,750.0	3,750.0	0.0	Exact	
15/9-F-15	4,090.0	4,105.0	15.0	Close	
15/9-F-15 A	4,095.0	4,095.0	0.0	Exact	
15/9-F-15 B	3,497.0	3,497.0	0.0	Exact	
15/9-F-15 C	3,232.0	3,232.0	0.0	Exact	
15/9-F-15 D	4,685.0	4,685.0	0.0	Exact	
15/9-F-4	3,510.0	3,510.0	0.0	Exact	
15/9-F-5	3,792.0	3,792.0	0.0	Exact	
15/9-F-7	1,085.0	1,085.0	0.0	Exact	
15/9-F-9	1,085.0	1,085.0	0.0	Exact	
15/9-F-9 A	1,206.0	1,206.0	0.0	Exact	

Exact: difference of 0.5 m or less.

Of the 26 wellbores, 24 have a defensible NOD counterpart, and 22 of these agree exactly on total depth. 15/9-F-15 differs by 15 m (0.4%) and 15/9-19 S by 227 m. Among the 19 F-series wellbores matched by literal name, 18 (94.7%) agree exactly, and 15/9-F-15 is again the exception. 15/9-F-7 and 15/9-F-9 share a total depth of 1,085 m, and NOD independently records 1,085 m for each, which rules out a duplication in the extraction.

One explanation covers the two unmatched wellbores and the three depth-based matches. Each WITSML technical sidetrack (19 BT2, 19 ST2, F-11 T2) has no separate NOD entry, and NOD records the final total depth of the sidetrack against the parent designation (19 B, 19 SR, F-11). The WITSML parent records that end short of that depth, 15/9-19 B at 3,272 m and 15/9-F-11 at 347 m, are the legs abandoned when the technical sidetrack was drilled, and they have no NOD counterpart. The stuck-pipe event of 15/9-19 B is documented in the narrative (Section 3.5), and the designation itself is the evidence for 15/9-F-11. The operator's WITSML system tracks each drilled leg while the national register records the wellbore. The two registers therefore differ by construction, and neither is in error.

The WITSML series of 15/9-19 S ends at 3,353 m, 227 m short of NOD's structured total depth (3,580.0 m). The well was an original exploration wellbore targeting the Theta Vest structure north of Sleipner East and planned as deviated, consistent with its "S" suffix. NOD's wellbore history records it as temporarily suspended at 3,569.0 m (2,394.0 m TVD RKB), immediately above the Heimdal sands (NOD 2026d), which is 216 m deeper than the last WITSML depth. The 11.0 m difference between the two NOD values lies within the register itself, and Table 5 uses the structured field for this wellbore as for every other. The archive holds no reports for the final drilling and suspension of 15/9-19 S, and the 227 m gap reflects those missing records. The reports it does hold are incomplete throughout, 39 reports over 91 days (Table 8), and the records do not show why.

5.5 Spud Date Validation

Spud date was compared across three sources for every wellbore, the WITSML spud_date field, free-text spud language in the earliest DDRs of the wellbore and the NOD entry date. The offset is the WITSML spud date minus the NOD pre-drill entry date where NOD records one, because it marks the start of top-hole drilling, and otherwise the NOD main entry date. Table 6 summarises the outcome and Fig. 5 plots the offset by category. For the four wellbores whose archive contains the NOD pre-drill entry (15/9-F-4, F-5, F-14 and F-15), the first report falls on that date, or one day later for F-4, which supports the pre-drill date as the spud reference. The categories were first assigned by inspection of the three sources for each wellbore, and the rules in the Rule column of Table 6 were formalised afterwards to describe those assignments. Applied mechanically, the rules reproduce 25 of the 26 assignments. The exception is 15/9-F-11, which was assigned Exact by inspection and is unresolved under the rule. NOD records a pre-drill campaign (6 to 17 April 2012) that is absent from the DDR archive, and the archive's 2013 reports describe a fresh spud from seabed, including the 30-inch conductor. The records do not show how the two campaigns relate.

Table 6: Spud date validation outcomes

Outcome	Wellbores	Rule	Detail
Exact agreement	6	R4: WITSML date within 1 day of the NOD date, and the first report or free-text spud follows it by at most 1 day.	15/9-19 A, F-10, F-15 A, F-15 B, F-15 C, F-15 D
Definitional offset	8	R6: WITSML date within 8 days of the NOD date, and the free-text operational start follows it by 2 to 20 days. R5: a parent leg whose NOD record is claimed by a technical sidetrack and that has no free-text spud date of its own (15/9-19 B).	Administrative spud precedes operational start by days to weeks (15/9-19 B, F-1 A, F-1 B, F-1 C, F-5, F-9 A, F-11 B, F-14)
WITSML data-entry error	4	R2: WITSML date differs from the first report or free-text spud by more than 30 days, while the NOD date agrees with them within 3 days.	15/9-F-4, F-7, F-12, F-15
NOD date marks later campaign	1	R3: WITSML date within 7 days of the first report, but more than 1 year from the NOD date.	15/9-F-1
Inherited parent spud date	2	R1: the sidetrack carries the same spud date as its parent.	15/9-F-11 A, F-11 T2

Outcome	Wellbores	Rule	Detail
Unresolved	5	R5: a technical sidetrack whose NOD record is its parent leg's (15/9-19 BT2 with 19 B, 19 ST2 with 19 SR). R7: any wellbore that no other rule fits.	15/9-19 BT2, 19 S, 19 ST2, F-9, F-11 (NOD records a pre-drill campaign, 6 to 17 April 2012, that is absent from the DDR archive; the 2013 reports describe a fresh spud from seabed, so the relationship between the campaigns cannot be determined from these records)
Total	26		

Rules are applied in the order R1 (inherited), R5 (technical sidetrack), R2 (error), R3 (later campaign), R4 (exact), R6 (definitional offset), R7 (unresolved). Unresolved includes 15/9-19 S, F-9 and F-11 (R7).

In the four error cases the structured spud_date differs from the first report and its narrative by 91 to 340 days, while the first report and the NOD date agree with each other. 15/9-F-12 is the clearest case. Its first report (14 June 2007) describes making up the 36 in. hole opener and agrees with the NOD entry date, and only the structured field (15 March 2007) disagrees. Two sidetracks, 15/9-F-11 A and 15/9-F-11 T2, carry a spud_date identical to that of their parent, which is almost certainly an inherited field value. NOD does not register 15/9-F-11 T2 separately, and Fig. 5 excludes it for that reason.

The spud date of 15/9-F-1 needed particular care. Its WITSML spud_date (28 November 2007) falls four days before its first DDR (2 December 2007), which narrates drilling 36 in. hole from the seabed, an ordinary gap between spud and first report. The 2007 reports cover the top hole only (four reports, maximum depth 226 m). The well was drilled to its total depth of 3,632 m in the 2013 campaign, whose first report (22 July 2013) is one day before the NOD entry date (23 July 2013). NOD records no pre-drill date for the 2007 work, unlike F-4, F-5, F-14 and F-15. The NOD date therefore marks the later campaign, and the case is classified as "NOD date marks later campaign". The WITSML value is consistent with the chronology of the wellbore.

5.6 Sidetrack Kick-Off Depth Recovery

Kick-off depth was recovered from the DDRs because the structured kick-off field is sparse. Kick-off here means the depth at which a sidetrack departs its parent wellbore, not the depth at which an original wellbore begins to build angle; the structured kick-off field of 15/9-19 S (562 m in all its reports) is consistent with the latter and is not used. One rule was applied to all sidetracks, and the depth was read from the last report of the parent and the first report of the sidetrack. The kick-off is the depth at which the narrative first reports new formation or departure from the parent hole. Where the narrative states none, it is the depth at which the kick-off or window was started, and for casing exits by a milled window it is the top of the window. Where the narrative is empty, it is the measured depth at the parent-to-child transition.

A kick-off depth was recovered for all 15 sidetracks (Table 7). The depth is written in the narrative for 11 of them. For 3 (15/9-F-1 A, F-1 B and F-15 D) the narrative names the kick-off event but gives no depth, and the depth is taken from the measured depth of the same report. For 15/9-19 ST2 both reports at the transition are empty. The depth is the measured depth there, which makes it the weakest case. The last report of F-1 B (22 February 2014) reads "Commenced time drilling to1364 m. 50% formation obtained at 1364 m.", and the first report of F-1 C drills 17½ in. hole from that depth, so the well was sidetracked at 1,364 m. The narratives report 3,059 m on 13 March 2014, but that depth marks the start of the 8½ in. section after casing was run. The sidetrack 15/9-19 A kicked off at the top of a milled window at 2,178 m (window milled from 2,178 to 2,183 m), a depth that NOD also records.

Table 7: Sidetrack kick-off depths, 15 sidetracks

Sidetrack	Parent	Kick-off depth (m MD)	Method	NOD kick-off (m)	NOD minus DDR (m)	Confidence
15/9-19 A	15/9-19 S (per NOD)	2,178	depth in text	2,178	0	HIGH
15/9-19 B	15/9-19 A	2,231	depth in text	2,200	-31	HIGH
15/9-19 BT2	15/9-19 B	2,911	depth in text			HIGH
15/9-19 ST2	15/9-19 S	1,536	depth sequence (narrative empty)			LOW
15/9-F-1 A	15/9-F-1	2,620	event in text, depth from md	2,620	0	MEDIUM
15/9-F-1 B	15/9-F-1 A	2,617	event in text, depth from md	2,604	-13	MEDIUM
15/9-F-1 C	15/9-F-1 B	1,364	depth in text	1,364	0	HIGH
15/9-F-9 A	15/9-F-9	421.5	depth in text	428	+6.5	HIGH
15/9-F-11 T2	15/9-F-11	246	depth in text			MEDIUM
15/9-F-11 A	15/9-F-11 T2	2,586	depth in text	2,578	-8	HIGH
15/9-F-11 B	15/9-F-11 A	2,589	depth in text	2,586	-3	LOW
15/9-F-15 A	15/9-F-15	1,381	depth in text	1,381	0	HIGH
15/9-F-15 B	15/9-F-15 A	3,185	depth in text	3,185	0	HIGH
15/9-F-15 C	15/9-F-15 B	2,566	depth in text	2,575	+9	MEDIUM
15/9-F-15 D	15/9-F-15 C	1,380	event in text, depth from md	1,380	0	MEDIUM

Depths are m MD. NOD minus DDR is NOD's wlbKickOffPoint minus the recovered depth; blank where NOD has no value. NOD lists 15/9-19 S as the parent; the re-entry from which 15/9-19 A was drilled (NOD 15/9-19 SR2, 1997) has no WITSML reports, so the kick-off is read from the 15/9-19 A reports.

No kick-off depth exists for 15/9-19 S. Its earliest narratives describe a top-hole spud, consistent with NOD's record of it as an original exploration wellbore.

NOD's wlbKickOffPoint agrees with the recovered depth to within 10 m for 10 of the 12 sidetracks for which it is available. It equals the recovered depth for 6 (15/9-19 A, F-1 A, F-1 C, F-15 A, F-15 B and F-15 D), differs by 10 m or less for 4 (15/9-F-9 A, F-11 A, F-11 B and F-15 C) and by more for 2 (15/9-19 B, 31 m; 15/9-F-1 B, 13 m). NOD has no kick-off value for the other 3 (15/9-19 BT2, 19 ST2 and F-11 T2). The least certain depths are 15/9-19 ST2, which rests on the measured depth alone, and 15/9-F-11 B, where cement and new formation were drilled from 2,531 to 2,589 m and the end of the interval is taken as the departure.

5.7 Duration to Total Depth

Table 8 gives the duration of each wellbore from its first report (the 1980 date sentinel excluded) to the date total depth was first reached, which is the maximum valid depth of Table 5. It lists the elapsed calendar days, the number of reports in that span (inclusive), the drilled length and the metres per report-day. The drilled length is total depth minus the kick-off depth of Table 7 for the 15 sidetracks and total depth for the other 11, and the rate divides it by the number of reports. Elapsed days and reports agree where a wellbore was reported daily and differ where the archive has gaps. Seven of the 26 wellbores, whose elapsed days exceed twice their reports, are flagged (15/9-19 S, F-1, F-4, F-5, F-9, F-14 and F-15). F-1 reaches its total depth of 3,632 m on 17 August 2013, 2,085 days after its first report on 2 December 2007, in 31 reports, four of them from 2007. The totals of 15/9-19 B (3,272 m) and 15/9-F-11 (347 m) are the depths of an abandoned parent leg or stub (Section 5.4) and make their durations indicative only. Drilled lengths and rates for sidetracks depend on the kick-off depths of Section 5.6. The rates are per reported day, and for F-1 they mix two campaigns. Per-phase report counts for each wellbore and well family are published with the duration table.

Table 8: Duration to total depth (TD) and drilled length, 26 wellbores

Wellbore	First report	TD (m)	TD reached	Elapsed days to TD	Reports to TD	Gap flag	Drilled length (m)	m per report-day
15/9-19 A	1997-07-25	4,131	1997-08-22	28	29		1,953	67.3
15/9-19 B (indicative)	1997-11-10	3,272	1997-11-28	18	19		1,041	54.8
15/9-19 BT2	1997-12-06	4,250	1998-01-11	36	37		1,339	36.2
15/9-19 S	1992-09-09	3,353	1992-12-09	91	39	yes	3,353	86.0
15/9-19 ST2	1992-12-19	4,641	1993-03-12	83	68		3,105	45.7
15/9-F-1	2007-12-02	3,632	2013-08-17	2,085	31	yes	3,632	117.2
15/9-F-1 A	2013-08-24	3,682	2013-08-26	2	3		1,062	354.0
15/9-F-1 B	2013-08-30	3,465	2013-09-07	8	8		848	106.0
15/9-F-1 C	2014-02-22	4,094	2014-03-15	21	22		2,730	124.1
15/9-F-10	2009-04-07	5,331	2009-06-03	57	58		5,331	91.9
15/9-F-11 (indicative)	2013-03-08	347	2013-03-20	12	13		347	26.7
15/9-F-11 A	2013-05-15	3,762	2013-05-22	7	7		1,176	168.0
15/9-F-11 B	2013-05-29	4,770	2013-06-12	14	15		2,181	145.4
15/9-F-11 T2	2013-03-24	4,562.1	2013-05-05	42	43		4,316.1	100.4
15/9-F-12	2007-06-14	3,520	2007-08-25	72	73		3,520	48.2
15/9-F-14	2007-11-06	3,750	2008-06-15	222	61	yes	3,750	61.5
15/9-F-15	2007-11-19	4,090	2008-11-29	376	55	yes	4,090	74.4
15/9-F-15 A	2008-12-12	4,095	2009-01-05	24	25		2,714	108.6
15/9-F-15 B	2009-01-14	3,497	2009-01-15	1	2		312	156.0
15/9-F-15 C	2009-01-19	3,232	2009-02-14	26	27		666	24.7
15/9-F-15 D	2013-11-11	4,685	2013-12-02	21	22		3,305	150.2
15/9-F-4	2007-06-10	3,510	2008-02-16	251	40	yes	3,510	87.8
15/9-F-5	2007-11-15	3,792	2008-07-30	258	31	yes	3,792	122.3
15/9-F-7	2007-09-15	1,085	2007-10-06	21	22		1,085	49.3
15/9-F-9	2007-11-29	1,085	2008-08-30	275	14	yes	1,085	77.5
15/9-F-9 A	2009-07-04	1,206	2009-07-09	5	6		784.5	130.8

Gap flag: elapsed days to TD exceed twice the reports to TD. Drilled length is total depth minus the Table 7 kick-off depth for sidetracks and total depth for original wellbores; for the original wellbores this is measured depth from the rig datum and includes the water depth. (indicative): TD is the depth of an abandoned parent leg or stub.

6 Discussion

6.1 What the Dataset Supports

The 1,421-row depth-time-phase dataset can be queried directly. Depth-versus-report-count curves (Fig. 3) allow wellbores drilled from the same platform to be compared, phase durations can be extracted for offset-well planning, and the classified phases locate casing, cementing and P&A operations in time and depth for each wellbore. Because the classifier is deterministic and the audit quantifies its residual, a user can bound how much of any phase-based statistic is missing. At the attainable-rate estimate, roughly 199 DDRs of construction content remain unassigned, mostly drilling progress reported without a hole size. Table 8 offers elapsed days, reports and metres per report-day

to total depth for each wellbore as a first benchmark, and the gap flag marks wellbores whose elapsed time cannot be read as drilling time.

6.2 Data-Quality Lessons for WITSML Archive Users

Five characteristics of this archive would mislead an analysis that took the structured fields at face value, and each is likely to recur in other WITSML archives.

- Placeholder values pass null tests. The depth sentinel '-999.99', the zero string '0', the string "None" (in the narrative and forecast fields) and the date sentinel affect 515 field values.
- A date sentinel (1 January 1980) shifts the entire drilling campaign of a wellbore into the wrong era of any time-based analysis if it is treated as a genuine first report.
- Structured spud dates contradict the first report of the same wellbore in four wellbores and are inherited from the parent in two sidetracks.
- The structured kick-off field is populated in only 14.10% of rows, which forces the recovery of sidetrack geometry from the narrative.
- Technical-sidetrack designations have no counterpart in the national register, and a join on wellbore name silently drops the valid records.

None of these invalidates the archive, but each needs an explicit rule, and the rules used here are published with the code.

6.3 Relevance to Records-Based Screening of Legacy Wellbores

Assessments of depleted fields for CO2 storage, and decommissioning reviews generally, rely on an inventory of existing wellbores and on evidence of how each was constructed and abandoned. The reconstruction shows that a national register alone gives an incomplete inventory in three ways. Abandoned parent legs below technical sidetracks (15/9-19 B to 3,272 m and 15/9-F-11 to 347 m) exist physically but have no register entry of their own. Wellbores that share a physical interval, such as the four wellbores of the 15/9-F-1 family, which share the hole above 1,364 m MD, are not independent penetrations above their kick-off depths. The archive can also stop, as it does for 15/9-19 S, which has no reports covering its final 227 m. A screening assessment should record such a gap as an evidence gap without inferring the missing history. The phase-classified timeline is a starting point for locating the casing, cementing and plugging records that such an assessment needs.

6.4 Transferability

The audit protocol, the placeholder checks and the wellbore-matching logic transfer directly to WITSML 1.4-series drillReport archives (tested here on 1.4.0.0). The pattern set is tuned to the English-language narratives and abbreviations of one operator over the life of one field, Applying it elsewhere requires a pattern review. The audit should be repeated to measure the new residual, since the Volve figure cannot be assumed to carry over.

7 Limitations

The classifier assigns a single primary phase to each DDR. A DDR that narrates both drilling progress and cementing takes the higher-priority drilling phase. The Cementing count therefore underrepresents the days on which cementing occurred alongside another operation.

The audit labelled construction content only. Unclassified records that describe operations belonging to non-construction phases of the classifier, for example waiting or rig moves, were labelled non-construction, which makes the attainable rate conservative with respect to those phases. The sample of 50 was labelled by a single reviewer, the author, and no inter-rater agreement was measured. The interval of 82.5 to 88.3% reflects the modest sample size.

Two sampled records sit in a group that does not match their operational context, although both were labelled correctly on content. The 1980 date sentinel of 15/9-19 A places the 1997 drilling of that wellbore in the production group, and the 2013 drilling campaign reports of 15/9-F-1 also fall into the production group.

The timeline includes only DDRs with a genuine measured depth and applies no interpolation. A discontinuity where a run of DDRs carries a placeholder value marks missing data and allows no conclusion about drilling activity in that period.

Kick-off depths were read from the narratives by one rule, and two of them rest on weaker evidence, 15/9-19 ST2 on the measured depth alone and 15/9-F-11 B on an interval of cement and new formation (2,531 to 2,589 m). NOD gives no kick-off point for three sidetracks. The four wellbores of the 15/9-F-1 family (F-1, F-1 A, F-1 B and F-1 C) share the hole above 1,364 m MD, and F-1, F-1 A and F-1 B share it above approximately 2,617 to 2,620 m MD. Any formation-level analysis that treats them as independent must account for this. Metres per report-day are rates per reported day, and gaps between reports in the archive affect the elapsed time to total depth (the gap flag in Table 8).

No isolated ablation was run for the effect of Unicode normalisation on the classification rate. The survey finds ½ in 61 DDRs and ¼ in 10, and the normalisation handles both, but its effect was not separated from other refinements of the pattern set.

The analysis starts from the raw XML, which the published code extracts with every field kept as a raw string, and an extraction error would still propagate to every result. The near-exact agreement with NOD on total depth (22 of 24) indicates that the depth fields were extracted faithfully.

8 Conclusions

This paper shows three things. First, a field's daily drilling reports can be turned into a construction history that is checked against the regulator's register and whose error is measured, using a method anyone can inspect and rerun. Second, the structured fields of a WITSML archive cannot be trusted on their own: spud dates, kick-off depths and placeholder values all needed the narrative or the national register to correct them, and a kick-off read naively from the narrative can pick up the start of a hole section instead of the point where the sidetrack leaves its parent (3,059 m against 1,364 m for 15/9-F-1 C). Third, the national register alone does not give a complete inventory of what was drilled: abandoned parent legs, shared upper hole and missing campaigns appear only when the reports are reconstructed. For anyone screening old wells for CO2 storage or decommissioning, the practical lesson is to reconstruct the well history from the reports before relying on either source alone.

[TJ: rewrite in your own words]

1. A deterministic, 13-phase rule-based classifier assigns 1,303 of 1,759 Volve WITSML DDRs (74.1%) to a construction phase, with every decision traceable to a published pattern.
2. A stratified audit finds that 52% of the sampled unclassified narratives describe construction operations, which gives an estimated attainable classification rate of 85.4% (95% CI 82.5 to 88.3%). The main gaps are drilling progress reported without a hole size, non-standard abbreviations, and fishing and fracturing operations.
3. Reconstructed total depth agrees exactly with the NOD register for 22 of 24 matched wellbores. The two differences, 15 m on 15/9-F-15 and a records-coverage gap of 227 m on 15/9-19 S, are both explained.
4. Every naming mismatch between the WITSML archive and the national register follows the Norwegian technical-sidetrack convention. Parent legs abandoned below technical sidetracks exist physically but have no register entry of their own.
5. The structured spud date needs checking against the first DDR. Four wellbores carry data-entry errors, two sidetracks carry inherited values, and one apparent error is a NOD entry date that marks a later drilling campaign.
6. Sidetrack kick-off depths were recovered for all 15 sidetracks by one rule. 15/9-F-1 C was sidetracked at 1,364 m, and 3,059 m marks the start of its 8½ in. section. The four wellbores of the 15/9-F-1 family share their hole above 1,364 m.
7. Placeholder values that pass a null test affect 515 field values in the archive and must be removed explicitly before analysis.
8. The resulting 1,421-row depth-time-phase dataset and the classifier are published openly.

Future work should enlarge the audit sample with a second independent labeller, close the identified pattern gaps, validate the reconstructed timelines against the Final Well Reports and End of Well Reports held in the Volve archive, and extend the duration results to phase-level durations across the campaign.

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Generative AI (Anthropic Claude, used through Claude Code) built and ran the reproducible analysis pipeline from the raw data and was used to assist with debugging, checking arithmetic and editing the manuscript text. The sidetrack kick-off depths were proposed by the AI tool from the narratives under one rule. The author rejected the tool's first proposals for three sidetracks (15/9-19 A, 15/9-F-9 A and 15/9-F-11 B) and kept the earlier values, then reviewed all 15 depths and approved every one without change. The author made all 50 audit labels, reviewed all AI-assisted content, and takes full responsibility for the paper.

Code and Data Availability

The classifier, the extraction and analysis code, the 1,421-row timeline dataset, the audit labels (file name, wellbore, date, stratum, call and miss reason), the NOD comparison tables, the kick-off and duration tables, and a numbers registry tracing every reported figure to the script that produced it are available at https://github.com/985185/volve-well-construction-timeline. DDR narrative text is not redistributed: the Volve data are available from Equinor under the Equinor Open Data Licence (https://www.equinor.com/energy/volve-data-sharing), and the data files of the repository hold only file names, dates, depths, categories and quotes of 15 words or fewer. The check against the author's earlier extraction (Section 5.1) is reported but is not needed to reproduce any figure.

Abbreviations

BHA = bottom-hole assembly
BOP = blowout preventer
CI = confidence interval
DDR = Daily Drilling Report
DP = drill pipe
EZSV = drillable cement retainer or bridge plug (Halliburton trade name)
LWD = logging while drilling
MD = measured depth
MT = metric tonnes
MWD = measurement while drilling
NOD = Norwegian Offshore Directorate (formerly Norwegian Petroleum Directorate)
P&A = plug and abandonment
POOH = pull out of hole
RKB = rotary kelly bushing
SG = specific gravity
TD = total depth
TVD = true vertical depth
VSP = vertical seismic profile
WITSML = Wellsite Information Transfer Standard Markup Language
WOW = waiting on weather

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Figures

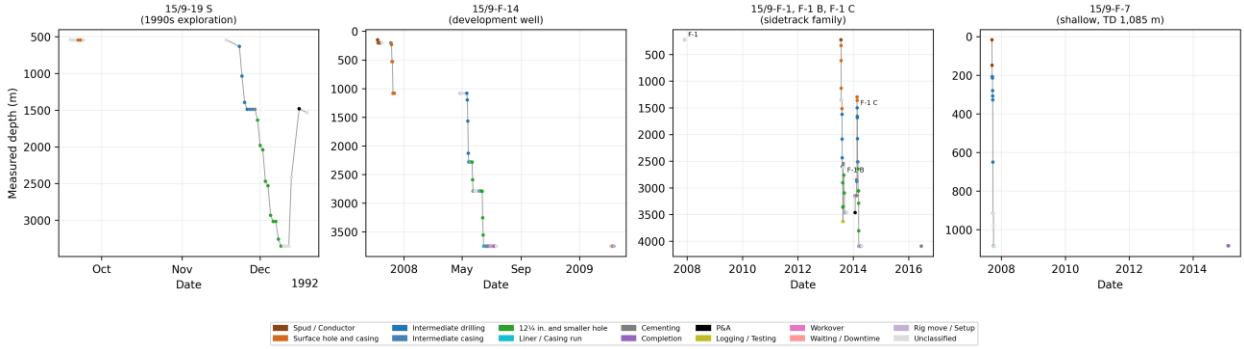


Fig. 2: Depth versus time for four representative wellbores, coloured by classified phase.

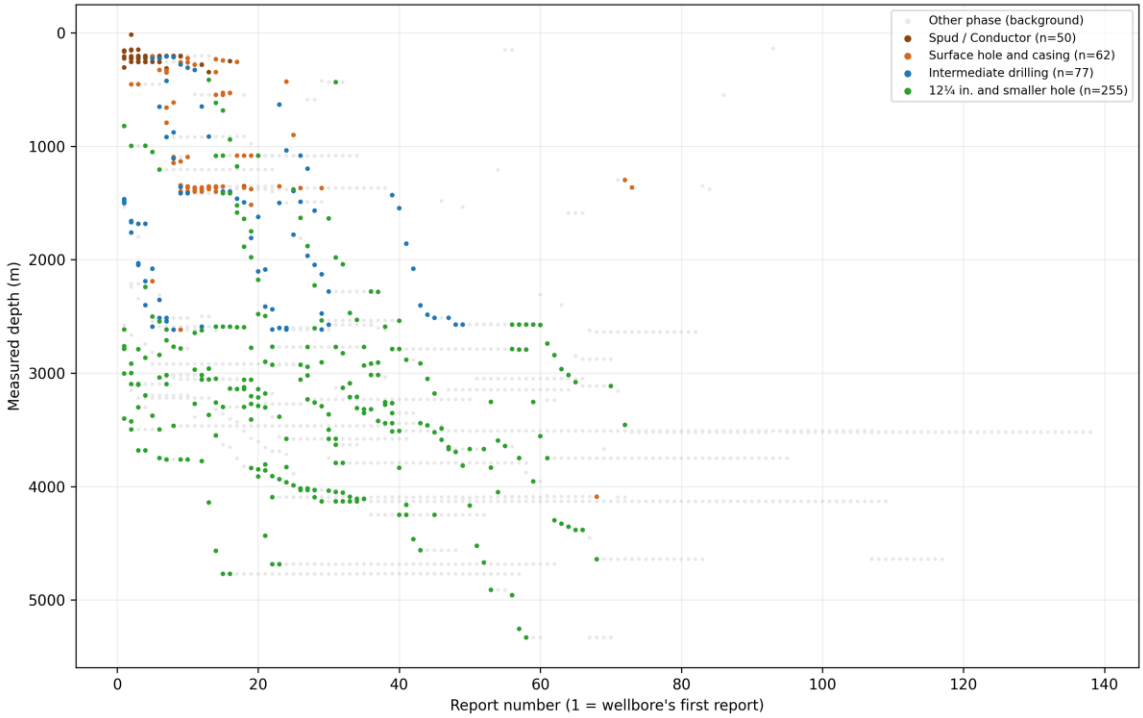


Fig. 3: Report number (1 = the wellbore's first report, as counted in Table 8) versus measured depth for all 26 wellbores, coloured by hole section (other phases and unclassified reports in grey).

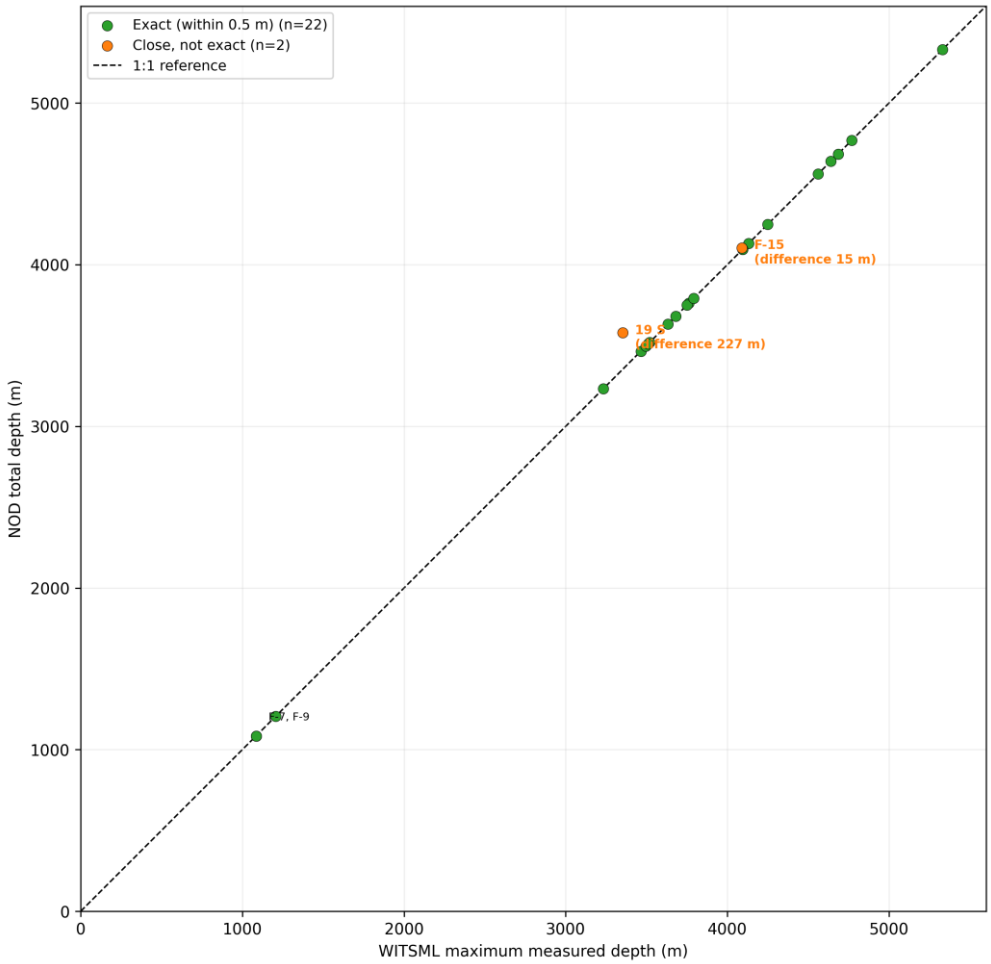


Fig. 4: Reconstructed WITSML maximum MD versus NOD total depth for the 24 matched wellbores, with 1:1 reference line. The two unmatched wellbores (15/9-19 B and 15/9-F-11) are not plotted.

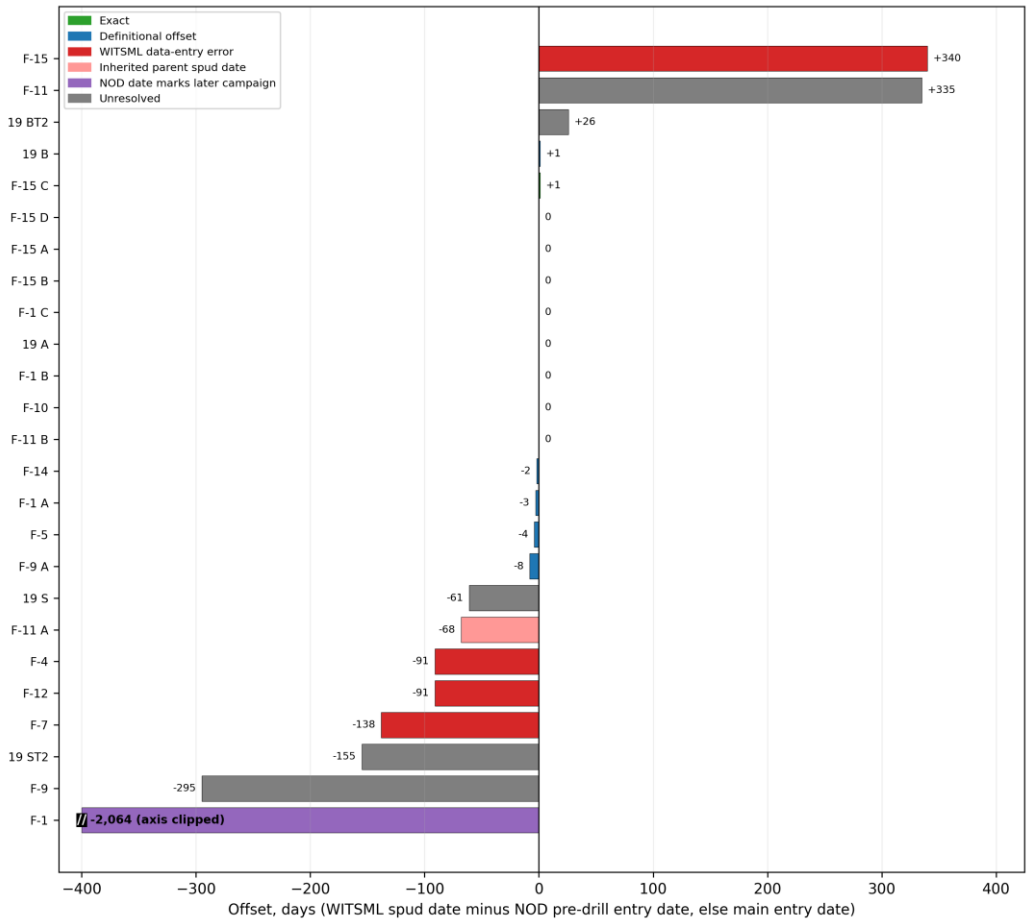


Fig. 5: Spud date offset (WITSML spud date minus the NOD pre-drill entry date where recorded, otherwise the NOD main entry date) for 25 wellbores, coloured by validation category. The bar of 15/9-F-1 (offset -2,064 days) is clipped at -400 days and labelled with its value. 15/9-F-11 T2 is excluded because NOD does not register it separately.

Appendix A: Traceability

Every quantitative figure in this paper traces to an entry in the numbers registry published in the repository (243 entries, each with the producing script and function, the SHA-256 of its input files and an execution timestamp). The audit estimates in Section 5.2 are derived arithmetically from registry entries, and the arithmetic is shown in Appendix B.

Appendix B: Audit Arithmetic

This appendix gives the arithmetic behind the attainable classification rate of Section 5.2.

The sample proportion is 26/50 = 0.520, with a Wilson-score 95% interval of 0.385 to 0.652. The sample comes from the 383 records with a genuine narrative among the 456 unclassified records, and the proportion is therefore scaled by 383/1,759 = 0.2177. The estimated residual miss is 0.520 × 0.2177 = 0.1132, or 11.3 percentage points (approximately 199 DDRs). Adding the observed rate (1,303/1,759 = 0.7408) gives an estimated attainable classification rate of 85.4%. Propagating the Wilson interval gives 0.7408 + 0.385 × 0.2177 = 0.825 and 0.7408 + 0.652 × 0.2177 = 0.883, a 95% interval of 82.5 to 88.3%. Weighting the two strata by their population shares (296/383 and 87/383) gives a sample proportion of 0.518 and an attainable rate of 85.3%. The result does not depend on whether the estimate is stratum-weighted.